

A machine-checked, uniform-in-order Abel formula for the
Casoratian,
with a formalized fundamental-system converse and an irrationality
certificate,
in Lean 4/Mathlib

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Abstract

For k sequences $s^{(0)}, \dots, s^{(k-1)}: \mathbb{N} \rightarrow R$ over a commutative ring R , all solving the order- k homogeneous linear recurrence $s(n+k) = \sum_{j=0}^{k-1} c_j(n) s(n+j)$, the *Casoratian* (discrete Wronskian) $C(n) = \det [s^{(t)}(n+i)]_{0 \leq i, t < k}$ obeys the first-order law $C(n+1) = (-1)^{k-1} c_0(n) C(n)$, hence the closed product form $C(n) = (\prod_{l < n} (-1)^{k-1} c_0(l)) C(0)$. Only the *lowest* coefficient c_0 survives; $c_1, \dots, c_{k-1}$ cancel. This is *Abel's formula* for the Casoratian — classical, going back to Casorati, the discrete analogue of Liouville's formula for the Wronskian. Our contribution is not the identity but its treatment: we give a single proof valid *uniformly in the order k* and formalize it in Lean 4/Mathlib to a clean axiom cone (a subset of `{propext, Classical.choice, Quot.sound}`, no `sorry`). The formal proof rewrites the last row of $C(n+1)$, via the recurrence, into a linear combination of the *cyclically rotated* rows of $C(n)$; `Matrix.det.updateRow.sum` collapses the combination to the single coefficient $c_0(n)$ and `sign.finRotate` supplies the cyclic sign $(-1)^{k-1}$. We also formalize the constant-coefficient power form $C(n) = ((-1)^{k-1} c_0)^n C(0)$, an integral-domain non-vanishing (linear-independence) certificate, and the order-2 and order-3 instances proved independently by hand as faithfulness witnesses (order 2 reproduces the corpus sign $-c_0$, with Euler's $(-1)^n$ convergent Wronskian as the case $c_0 \equiv 1$; order 3 gives $+c_0$). The result opens the order- ≥ 3 direction in a formalized polynomial-continued-fraction corpus previously confined to the order-2 (three-term) case — the algebraic backbone of simultaneous / Hermite–Padé approximation and of Apéry-type irrationality certificates. Building on the law we further formalize its *fundamental-system converse* — forward determinacy of a solution by its first k values, and completeness of the basis once the initial Casoratian is a unit — so that the solutions form a machine-checked *fundamental set*; and we put that set to work through a reusable, machine-checked *irrationality pipeline* that turns any integer fundamental set together with a shrinking *real* Hermite–Padé remainder into a proof of irrationality, the Casoratian discharging the non-vanishing (linear-independence) input *structurally*. We instantiate it end-to-end on the order-3 recurrence $u(n+3) = 2u(n+2) - u(n)$ (characteristic roots $\varphi, \psi, 1$), whose fundamental set $\{1, F_n, F_{n+1}\}$ and remainder $F_n \varphi - F_{n+1} = -\psi^n \rightarrow 0$ re-derive the irrationality of the golden ratio φ through the Casoratian — an independent route from Mathlib's $\sqrt{5}$ -based proof, and a demonstration that the determinantal machinery genuinely closes an irrationality theorem. We claim no novelty for φ 's irrationality itself, which is classical; the contributions are the uniform-in- k machine-checked law, the formalized converse, and the reusable certificate engine. An independent computer-algebra gate confirms the identity symbolically at $k = 3$, by exact-rational checks for $k = 2, \dots, 7$, and on named instances.

2020 MSC: 39A06 (primary), 68V20, 11J72, 15A15. **Keywords:** Casoratian, discrete Wronskian, Abel’s formula, linear difference equation, higher-order recurrence, fundamental system, Hermite–Padé approximation, Apéry-type irrationality, irrationality criterion, golden ratio, Lean 4, Mathlib, formal verification, polynomial continued fractions.

1 Introduction

The convergents of a polynomial continued fraction $K_{n \geq 1} a_n/b_n$ are governed by a three-term recurrence $q_n = b_n q_{n-1} + a_n q_{n-2}$, and the basic structural invariant is the order-2 Casoratian (discrete Wronskian) of two solutions: it flips sign and scales by the lower coefficient at each step, so the convergent Wronskian $p_{n+1}q_n - p_n q_{n+1} = (-1)^n$ is constant up to sign. This is the certificate behind Euler’s irrationality criterion, and across the deposited corpus the Casoratian appears only in this order-2, three-term form ([12, 11]; formalized as `casoratian_eqG` in the companion `pcf-delta` project [13]).

The order- ≥ 3 direction. The natural next object is the Casoratian of $k \geq 3$ solutions of an order- k recurrence

$$s(n+k) = \sum_{j=0}^{k-1} c_j(n) s(n+j), \quad c_0, \dots, c_{k-1}: \mathbb{N} \rightarrow R, \quad (1)$$

the $k \times k$ determinant

$$C(n) = \det [s^{(t)}(n+i)]_{0 \leq i, t < k}. \quad (2)$$

This is the algebraic backbone of *simultaneous* (Hermite–Padé) approximation, the engine of Apéry-type irrationality proofs ([4]): there one builds several sequences satisfying one common higher-order recurrence and certifies their linear independence through the non-vanishing of exactly this determinant. The first-order law that C obeys is classical — the discrete analogue of Abel’s identity for the Wronskian of a linear ODE, going back to Casorati; see, e.g., [1, Ch. 2] and [2]. It states that C is reproduced at each step by a single scalar built only from the *lowest* coefficient c_0 .

Contribution. The identity itself is not new. What we provide is:

- (i) a single proof valid *uniformly in the order k* (Theorem 2), and its formalization in Lean 4 with Mathlib to a clean axiom cone (Section 7), so the result is `PROVEN` in the strict sense of this program: `#print axioms` reports a subset of `{propext, Classical.choice, Quot.sound}` with no `sorryAx` and no `sorry/admit/native_decide` in source;
- (ii) the closed product form, the constant-coefficient power form, and an integral-domain non-vanishing (linear-independence) certificate, all formalized (Section 4);
- (iii) the order-2 and order-3 instances proved *independently* by a direct cofactor computation, kept as faithfulness witnesses against the general proof and machine-checked to coincide with the general determinant (Proposition 6); and
- (iv) an independent computer-algebra gate (Section 8) that re-derives the law symbolically and on named instances, in the program’s “gate-before-Lean” discipline;

- (v) the *fundamental-system converse* (Section 5), formalized: forward determinacy of a solution by its first k values and completeness of the basis once the initial Casoratian is a unit, so that the solutions form a machine-checked fundamental set — the object Hermite–Padé / Apéry applications consume; and
- (vi) a reusable, machine-checked *irrationality pipeline* (Section 6) that converts any integer fundamental set plus a shrinking real Hermite–Padé remainder into a proof of irrationality — with the non-vanishing input discharged structurally by the Casoratian — instantiated end-to-end on a concrete order-3 recurrence to re-derive the irrationality of the golden ratio φ through the discrete Wronskian.

Concretely, this opens the order- ≥ 3 space inside a formalized PCF corpus that had treated only order 2, supplies a reusable, machine-checked independence certificate for future Hermite–Padé constructions, and — through the converse and the irrationality pipeline — carries that certificate all the way to a worked irrationality theorem.

2 The Casoratian of a linear recurrence

Throughout, R is a commutative ring and $k \geq 1$ a fixed order. We write a solution of (1) simply as a sequence $s: \mathbb{N} \rightarrow R$ satisfying that identity for all n ; no invertibility of the coefficients is assumed.

Definition 1 (Casoratian matrix and Casoratian). *For sequences $s^{(0)}, \dots, s^{(k-1)}: \mathbb{N} \rightarrow R$, the Casoratian matrix $W(n)$ at n is the $k \times k$ matrix whose (i, t) entry is the i -shift of the t -th sequence, $s^{(t)}(n + i)$. The Casoratian is its determinant, $C(n) = \det W(n)$, the scalar of (2). Throughout we write W for the matrix and reserve $C = \det W$ for the scalar determinant; in the formalization the matrix is `casoMat` and the Casoratian is `(casoMat s n).det`.*

Because $\det$ is invariant under transposition, it is immaterial whether one indexes rows by shifts and columns by solutions or vice versa; we fix rows = shifts. The two “ends” of the next matrix $W(n + 1)$ are what drive the law:

- **Top $k - 1$ rows.** For $0 \leq i \leq k - 2$, row i of $W(n + 1)$ is the shift vector $(s^{(i)}(n + 1 + i))_t$, which is precisely row $i + 1$ of $W(n)$. So the top $k - 1$ rows of $W(n + 1)$ are the bottom $k - 1$ rows of $W(n)$, in order.
- **Bottom row.** Row $k - 1$ of $W(n + 1)$ is the shift vector at $n + k$, $(s^{(k-1)}(n + k))_t$, which by the recurrence (1) equals $\sum_{j=0}^{k-1} c_j(n) (s^{(j)}(n + j))_t$, a linear combination of *all* k rows of $W(n)$.

3 The general-order Abel formula for the Casoratian

Theorem 2 (Abel’s formula for the Casoratian, uniform in k). *Let $s^{(0)}, \dots, s^{(k-1)}: \mathbb{N} \rightarrow R$ all satisfy the order- k recurrence (1). Then the Casoratian (2) satisfies the first-order law*

$$C(n + 1) = (-1)^{k-1} c_0(n) C(n) \quad (n \in \mathbb{N}), \quad (3)$$

and consequently the closed product form

$$C(n) = \left(\prod_{l=0}^{n-1} (-1)^{k-1} c_0(l) \right) C(0) = (-1)^{(k-1)n} \left(\prod_{l=0}^{n-1} c_0(l) \right) C(0). \quad (4)$$

Only the lowest coefficient c_0 enters; $c_1, \dots, c_{k-1}$ cancel.

Proof. Let ρ be the cyclic permutation of $\{0, 1, \dots, k-1\}$ that moves every index up by one and wraps the top to the bottom, $\rho(i) = i+1$ for $i < k-1$ and $\rho(k-1) = 0$; it is a single k -cycle, so $\text{sgn } \rho = (-1)^{k-1}$. Let M be $W(n)$ with its rows permuted by ρ , i.e. row i of M is row $\rho(i)$ of $W(n)$. By the observation in Section 2, the top $k-1$ rows of M (namely rows $1, \dots, k-1$ of $W(n)$) are exactly the top $k-1$ rows of $W(n+1)$, while the bottom row of M is row 0 of $W(n)$.

Replacing the bottom row of M by the linear combination $\sum_{j=0}^{k-1} c_j(n)$ (row j of $W(n)$) produces exactly $W(n+1)$, because that combination is the bottom row of $W(n+1)$ and the top rows already agree. Now take determinants. Multilinearity in the replaced row — the content of `Matrix.det_updateRow_sum` — gives

$$\det W(n+1) = \sum_{j=0}^{k-1} c_j(n) \det(M \text{ with bottom row} = \text{row } j \text{ of } W(n)).$$

Every summand with $j \geq 1$ places into the bottom position a row of $W(n)$ that already occurs among the top $k-1$ rows of M , so that determinant vanishes. Only $j = 0$ survives, and there the bottom row is the original row 0 of $W(n)$, so the summand is $c_0(n) \det M$. Finally $\det M = \text{sgn } \rho \cdot \det W(n) = (-1)^{k-1} \det W(n)$. Recalling $C = \det W$, combining gives (3). A trivial induction on n , using $C(0)$ as the base case and (3) at the inductive step, yields the product form (4). $\square$

Remark 3 (What the formalization does, verbatim). *The Lean development (Section 7) realizes this proof literally. It builds M as `(casoMat s n).submatrix (finRotate (k+1)) id` (Mathlib’s `finRotate` is the cycle ρ), expresses $W(n+1)$ as that submatrix with its last row updated to the recurrence combination, then rewrites with `Matrix.det_updateRow_sum` (the multilinear collapse to the surviving coefficient) and `Matrix.det_permute` together with `sign_finRotate` (the cyclic sign). The “vanishing for $j \geq 1$ ” step is automatic: it is exactly the statement that `det_updateRow_sum` isolates the coefficient indexed by the updated row. No analytic input is used; the argument is finitary algebra over an arbitrary `CommRing`.*

4 Specializations and consequences

Corollary 4 (Constant lowest coefficient). *If $c_0(l) = a$ for all l , then $C(n) = ((-1)^{k-1}a)^n C(0)$. In particular, if a is a unit (or 1), the Casoratian never vanishes once $C(0) \neq 0$.*

Corollary 5 (Non-vanishing / linear-independence certificate). *Suppose R is an integral domain. If $C(0) \neq 0$ and $c_0(l) \neq 0$ for every $l < n$, then $C(n) \neq 0$. Hence if $C(0) \neq 0$ and c_0 is nowhere zero, the k solutions stay linearly independent at every window: $C(n) \neq 0$ for all n .*

Proof. Immediate from (4): a product of nonzero elements of a domain is nonzero, and $(-1)^{k-1}$ is a unit. $\square$

Corollary 5 is the order- k analogue of the order-2 statement $p_{n+1}q_n - p_nq_{n+1} = (-1)^n \neq 0$ that underlies Euler’s criterion: it is precisely the determinantal independence certificate one needs for Hermite–Padé systems.

Proposition 6 (Low-order witnesses). *The cases $k = 2, 3$ hold and are proved independently by direct cofactor expansion, without the permutation argument:*

$$k = 2: \quad C(n+1) = -c_0(n) C(n); \quad k = 3: \quad C(n+1) = +c_0(n) C(n).$$

The order-2 sign $-c_0$ is the corpus sign; its $c_0 \equiv 1$ instance is the constant convergent Wronskian $p_{n+1}q_n - p_nq_{n+1} = (-1)^n$.

These hand proofs (`caso2_step`, `caso3_step`; the 3×3 cofactor expansion is closed by `ring`) are retained as *faithfulness witnesses*: they agree with Theorem 2 at $k = 2, 3$, guarding against a vacuous or mis-signed general statement. Moreover, the bespoke order-3 expansion is reconciled with the general object by the machine-checked bridge `caso3_matches_general`, which proves `caso3 y z w n` equal to $\det W(n)$ for the general Casoratian matrix `casoMat` (Section 3) specialized to $k = 2$ with $\{y, z, w\}$; so the witnesses certify the *formalized* general theorem (Theorem 12), not merely a parallel polynomial identity.

5 The fundamental-system converse

The Abel law and its non-vanishing corollary say that the chosen solutions stay *independent*; the converse direction — that they are also *complete*, i.e. span every solution — is what upgrades $\{s^{(t)}\}$ from an independent family to a genuine *fundamental set*, the object Hermite–Padé and Apéry constructions actually consume. We formalize it in two parts.

Theorem 7 (Fundamental-system converse; PROVEN). *Let $s^{(0)}, \dots, s^{(k-1)} : \mathbb{N} \rightarrow R$ solve the order- k recurrence (1).*

- (a) (Forward determinacy.) *Any solution of (1) is determined by its first k values: if $y, z : \mathbb{N} \rightarrow R$ both satisfy (1) and $y(i) = z(i)$ for $0 \leq i < k$, then $y(n) = z(n)$ for all n . No invertibility hypothesis is needed.*
- (b) (Completeness.) *If the initial Casoratian $C(0)$ is a unit of R , then every solution y of (1) is an R -linear combination of the $s^{(t)}$: there is $a : \{0, \dots, k-1\} \rightarrow R$ with $y(n) = \sum_t a_t s^{(t)}(n)$ for all n , given explicitly by the coordinates $a = W(0)^{-1} (y(0), \dots, y(k-1))^T$.*

Together with the non-vanishing certificate (Corollary 5), $\{s^{(t)}\}$ is then a machine-checked fundamental set.

Proof (as formalized). Part (a) is a strong induction on n : for $n < k$ the hypothesis gives $y(n) = z(n)$ directly, and for $n \geq k$ the recurrence writes both $y(n)$ and $z(n)$ as the *same* R -combination of strictly earlier values, equal by the inductive hypothesis. For part (b), set $a = W(0)^{-1} (y(i))_{i < k}$ — legitimate because $C(0)$ is a unit, so the matrix $W(0)$ is invertible — so that $\sum_t a_t s^{(t)}(i) = y(i)$ on the first window $0 \leq i < k$. The combination $\sum_t a_t s^{(t)}$ is itself a solution of (1), so by part (a) it agrees with y at *every* n . In particular completeness needs only that $C(0)$ is a unit: the forward determinacy of (a) carries the initial-window agreement to all n with no reversibility — no hypothesis on c_0 — required, a small simplification over the two-hypothesis form in which we previously recorded this converse as merely STRUCTURAL. $\square$

Remark 8 (Lean). *Part (a) is `PcfGeneralCaso.sol_unique_of_init` (strong induction, `Nat.strong_induction_on`) part (b) is `PcfGeneralCaso.solution_isLinearCombo`, which takes coordinates from `Mathlib's nonsingular inverse` (`Matrix.mul_nonsing_inv` at the initial window) and then invokes part (a) to propagate. Both are in `GeneralCaso.lean` (order $k+1$ via `Fin (k+1)`, Remark 11), with cone `{propxt, Classical.choice, Quot.sound}`.*

6 An irrationality certificate from the fundamental set

We now put the fundamental set to number-theoretic work. The construction is the discrete heart of the Hermite–Padé / Apéry mechanism, reduced to its determinantal skeleton and machine-checked end to end.

The window is a Casoratian matrix–vector product. For coefficients $a: \{0, \dots, k-1\} \rightarrow R$, the linear form $L(n) = \sum_t a_t s^{(t)}(n)$ evaluated on the consecutive window $n, \dots, n+k-1$ is exactly $W(n) a$,

$$(L(n+i))_{0 \leq i < k} = W(n) a \quad (5)$$

(`caso_mulVec_apply`). Hence, over an integral domain, if $C(n) \neq 0$ the matrix $W(n)$ is injective and a *nontrivial* form ($a \neq 0$) cannot vanish on a full window of k consecutive indices (`linForm_window_ne_zero`); consequently such a form is *frequently nonzero* as $n \rightarrow \infty$ (`linForm_frequently_ne_zero`). This is precisely the non-degeneracy the irrationality machinery needs — and the non-vanishing Casoratian supplies it for free.

The criterion. The standard linear-form irrationality criterion is `irrational_of_linearForm_frequently`: for $\alpha \in \mathbb{R}$ and integer sequences p, q , if the form $q(n)\alpha - p(n)$ is frequently nonzero and tends to 0, then α is irrational. Were $\alpha = a/b$ rational ($b \neq 0$), then $b(q(n)\alpha - p(n)) = q(n)a - bp(n) \in \mathbb{Z}$ is a nonzero integer whenever the form is nonzero, so $|q(n)\alpha - p(n)| \geq 1/|b| > 0$ frequently, contradicting convergence to 0.

The capstone. `irrational_of_fundamentalSet_form` combines the two. Given an integer fundamental set s (so $C(n) \neq 0$ for all n) and a *real* coefficient vector $v \neq 0$ for which the remainder

$$q(n)\alpha - p(n) = \sum_t v_t s^{(t)}(n)$$

tends to 0, the conclusion is that α is irrational. The frequently-nonzero hypothesis of the criterion is discharged *structurally*: the nonzero integer Casoratian, cast into $\mathbb{R}$ (`Int.cast_det` and injectivity of $\mathbb{Z} \hookrightarrow \mathbb{R}$), stays nonzero, so by (5) the real remainder is frequently nonzero. The only genuinely analytic input is the limit, kept as a single labelled hypothesis in the program’s established style (cf. the conditional limits of the corpus’s Topic 3 and Topic 5).

Remark 9 (Why the coefficients must be real). *The vector v is real, not integer, and this is essential rather than cosmetic. With an integer form, the remainder $q(n)\alpha - p(n)$ would be forced to be a cast integer, which cannot shrink to 0 while remaining frequently nonzero — the criterion’s two hypotheses would be jointly unsatisfiable and the “certificate” vacuous. The honest Hermite–Padé remainder is a genuine non-integer real combination of integer solutions: the integer Casoratian certifies their independence, while the real coefficients let the remainder converge.*

A concrete instance: the golden ratio. Take the order-3 recurrence

$$u(n+3) = 2u(n+2) - u(n), \quad x^3 - 2x^2 + 1 = (x^2 - x - 1)(x - 1), \quad (6)$$

with characteristic roots $\varphi = \frac{1+\sqrt{5}}{2}$, $\psi = \frac{1-\sqrt{5}}{2}$, and 1. The three integer sequences $\{1, F_n, F_{n+1}\}$ — the constant solution and two shifts of Fibonacci — solve (6) (`fibSol_rec`), and their initial Casoratian is the Cassini determinant

$$C(0) = \det \begin{pmatrix} 1 & 0 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 2 \end{pmatrix} = 1$$

(`fibCaso0`); since here $c_0 \equiv -1$ is a unit, Corollary 5 gives $C(n) \neq 0$ for all n (`fibCaso_ne_zero`), so $\{1, F_n, F_{n+1}\}$ is a fundamental set. Because $|\psi| < 1$, the Hermite–Padé remainder shrinks,

$$F_n \varphi - F_{n+1} = -\psi^n \longrightarrow 0$$

(Binet’s identity, Mathlib’s `Real.fib_succ_sub_goldenRatio_mul_fib; fib_remainder_tendsto`). Feeding this through the capstone with the real coefficient vector $v = (0, \varphi, -1)$ yields

`goldenRatio_irrational_via_caso` : φ is irrational,

derived entirely through the discrete-Wronskian machinery.

Note that $v = (0, \varphi, -1)$ assigns coefficient 0 to the constant solution, so the form $F_n\varphi - F_{n+1}$ is two-term and this is deliberately the *minimal* instance — the 2×2 Casoratian of $\{F_n, F_{n+1}\}$ would already certify the non-vanishing here. What the order-3 pipeline genuinely exercises is its *machinery*: the 3×3 Casoratian matrix, its integer determinant cast into $\mathbb{R}$, and the window/criterion plumbing, all run at $k = 2$ (order 3). We keep the instance minimal on purpose, reserving a genuinely order-3-essential target — one in which all three solutions carry weight — for the open problem of Section 9.

Remark 10 (Honest status of this instance). *We claim no novelty for the irrationality of φ : it is classical and already in Mathlib as `Real.goldenRatio_irrational` (a $\sqrt{5}$ -based argument). Our `goldenRatio_irrational_via_caso` is an independent route: the irrationality argument does not invoke the irrationality of $\sqrt{5}$ — it performs no $\sqrt{5}$ -descent, deriving irrationality instead from the order-3 Casoratian fundamental set and a shrinking remainder through the linear-form criterion. (The roots φ, ψ are of course defined via $\sqrt{5}$, and $|\psi| < 1$ is a $\sqrt{5}$ fact; what differs is the mechanism of the proof, not the avoidance of $\sqrt{5}$ as a quantity.) It serves as a faithfulness/demonstration witness that the pipeline genuinely closes an irrationality theorem end to end, exactly as the by-hand low-order witnesses (Proposition 6) guard the general law. A geometric companion $\{1, 2^n, 3^n\}$ of $u(n+3) = 6u(n+2) - 11u(n+1) + 6u(n)$, with explicit Casoratian $C(n) = 2 \cdot 6^n$ (`gcaso_ne_zero`) and window certificate `geom_window_ne_zero`, exhibits the non-vanishing half on a second recurrence but carries no shrinking remainder (all its roots have modulus ≥ 1), so it is deliberately not promoted to an irrationality certificate. What remains genuinely open (Section 9) is to drive the same pipeline to a target not already known to be irrational.*

7 The machine-checked formalization

The development is in Lean 4 with Mathlib (toolchain pinned `leanprover/lean4:v4.30.0`, Mathlib `rev=v4.30.0`). It is built and axiom-checked inside the companion `pcf-delta` `PcfContinuant` project [13], whose `Check.lean` gate runs `#print axioms` on every declaration; the three source files `GeneralCaso.lean` (the uniform-in- k core and the fundamental-system converse, namespace `PcfGeneralCaso`), `HigherCaso.lean` (the by-hand low-order witnesses, namespace `PcfHigherCaso`), and `HermitePade.lean` (the irrationality pipeline and the concrete certificates, namespace `PcfHermitePade`) are mirrored in this catalogue under `verify/`.

Remark 11 (Index convention). *The formalization parametrizes the order as $k+1$ via `Fin (k+1)` (so the recurrence is $s(n + (k+1)) = \sum_{j \leq k} c_j(n)s(n+j)$ and the matrix is $(k+1) \times (k+1)$); its sign therefore reads $(-1)^k$. This is the prose order- k law (3) under $k \mapsto k+1$: an index shift only, with $(-1)^{(k+1)-1} = (-1)^k$.*

Theorem 12 (Formalized core; PROVEN). *The following twenty-six declarations are machine-checked with clean axiom cones — each a subset of $\{\text{propext}, \text{Classical.choice}, \text{Quot.sound}\}$, no `sorryAx`, sources free of `sorry/admit/native_decide`, `lake build` exit 0.*

- Uniform-in- k law (`PcfGeneralCaso`, `GeneralCaso.lean`): `casoMat_det_step` is the step law (3) $C(n+1) = (-1)^k c_0(n)C(n)$ (order $k+1$; Remark 11); `casoMat_det_eq` is the closed

product form (4); `casoMat_det_eq_const` is the constant-coefficient power form (Corollary 4); `casoMat_det_ne_zero_of_init` is the integral-domain non-vanishing certificate (Corollary 5).

- Fundamental-system converse (`PcfGeneralCaso`, `GeneralCaso.lean`): `sol_unique_of_init` (forward determinacy) and `solution_isLinearCombo` (completeness) — Theorem 7.
- Low-order witnesses and the bridge (`PcfHigherCaso`, `HigherCaso.lean`): `caso2_step` (the order-2 sign $-c_0$), `caso3_step` (the order-3 step $+c_0$), `caso3_eq` (its product form), `caso3_eq_const` (its constant-coefficient case), `caso3_ne_zero_of_init` (its domain non-vanishing certificate), and `caso3_matches_general` (the bespoke 3×3 expansion equals $\det W$ of the general `casoMat` at $k = 2$, reconciling the witnesses with the general theorem) — Proposition 6.
- Irrationality pipeline (`PcfHermitePade`, `HermitePade.lean`): `caso_mulVec_apply` (`window = W(n)a`, (5)), `linForm_window_ne_zero` (no full-window vanishing over a domain), `linForm_frequently_ne_zero` (frequent non-vanishing), `irrational_of_linearForm_frequently` (the linear-form irrationality criterion), and the capstone `irrational_of_fundamentalSet_form` — Section 6.
- Concrete order-3 certificates (`PcfHermitePade`, `HermitePade.lean`): the geometric witness `gsol_rec`, `gcaso0`, `gcaso_ne_zero` ($C(n) = 2 \cdot 6^n$), `geom_window_ne_zero`; and the golden-ratio certificate `fibSol_rec`, `fibCaso0` (Cassini $C(0) = 1$), `fibCaso_ne_zero`, `fib_remainder_tendsto` ($F_n\varphi - F_{n+1} = -\psi^n \rightarrow 0$), and `goldenRatio_irrational_via_caso` (φ irrational).

Of the twenty-six, exactly two close with the minimal cone `{propext, Quot.sound}` — the by-hand witnesses `caso2_step` and `caso3_step`; the remaining twenty-four additionally use `Classical.choice`. This `Classical.choice` usage is inherited from Mathlib’s general infrastructure — the determinant and permutation machinery (`Matrix.det`) throughout (including the determinant side of the bridge `caso3_matches_general`), the nonsingular matrix inverse (`Matrix.mul_nonsing_inv`) in the converse, and the real-analysis limit layer in the certificate — not a sign that any proof is in a meaningful sense less constructive than the by-hand cases; we report the cones factually rather than rank them. No proof introduces any axiom outside the three-element clean set.

This follows the program’s pattern: the law and converse have no labelled analytic hypothesis at all, being finitary algebra over a `CommRing` (the domain and completeness results add only `IsDomainR`, resp. a unit Casoratian); the irrationality certificate isolates its *single* analytic input — the limit $q(n)\alpha - p(n) \rightarrow 0$ — as a labelled hypothesis, discharged for the golden ratio by Mathlib’s Binet identity. The Casoratian is defined as a genuine `Matrix` determinant, so the result is the honest determinantal identity, not a bespoke $k = 2, 3$ polynomial expression — and the order-3 by-hand expansion is itself certified equal to that determinant by `caso3_matches_general` (Proposition 6). Only the order-2 *step* sign is recorded here, as `caso2_step`: the full order-2 closed-form suite already exists in `pcf-delta` as `casoratian_eqG` [13] and is not duplicated, whereas order 3 carries the full suite as the smallest genuinely higher-order witness.

8 Independent symbolic verification

In keeping with the catalogue’s gate-before-Lean discipline, a `sympy` harness (`harness_caso_k.py`) re-derives the law independently of the Lean proof:

- a fully *symbolic* proof at $k = 3$: with generic symbolic data and generic c_0, c_1, c_2 , $C(1) - c_0 C(0)$ expands to 0 and $C(1)$ is free of c_1, c_2 (only c_0 enters);
- *exact-rational* checks of both the step law (3) and the product form (4) for $k = 2, \dots, 7$, over integer-coefficient recurrences and initial data drawn from a *fixed deterministic seed* (`random.Random(7k+101t+5)` for trial t), so the runs are exactly reproducible, confirming the alternating sign $(-1)^{k-1}$; and
- named instances: the Tribonacci recurrence $s(n+3) = s(n+2) + s(n+1) + s(n)$ ($c_0 \equiv 1$) has a *constant* Casoratian, and, at $k = 3$ (where the sign $(-1)^{k-1} = +1$), the recurrence with $c_0(n) = n + 1$ gives $C(n) = n! C(0)$ — both matching (4) exactly.

All checks pass. The harness and the Lean sources are listed in Section 11.

9 Epistemic status

- **PROVEN** (Lean, clean cones; Theorem 12): the uniform-in- k step law and product form, the constant-coefficient power form, the integral-domain non-vanishing certificate, and the order-2/order-3 witnesses — no analytic input.
- **PROVEN** (Lean, clean cones; Theorem 7): the *fundamental-system converse* — forward determinacy (`sol_unique_of_init`) with no invertibility hypothesis, and completeness (`solution_isLinearCombo`) once $C(0)$ is a unit. This was recorded as `STRUCTURAL` in v1.0 and is now formalized in the uniform-in- k , time-varying setting. The formalization in fact *simplifies* the hypotheses we had anticipated: because forward determinacy needs no reversibility, completeness follows from the single assumption that $C(0)$ is a unit, with no separate hypothesis that c_0 be a unit. (Over a field this is the textbook fundamental-set theorem, [1, Ch. 2]; Mathlib’s `LinearRecurrence` encodes only the constant-coefficient case [6].)
- **PROVEN** (Lean, clean cones; Section 6): the *irrationality pipeline* — the window identity (5), the domain-window non-vanishing, the linear-form irrationality criterion, and the capstone `irrational_of_fundamentalSet_form` — together with its end-to-end instantiation `goldenRatio_irrational_via_caso` re-deriving φ irrational through the order-3 Casoratian. The certificate carries exactly one labelled analytic hypothesis (the remainder’s limit), discharged here by Mathlib’s Binet identity. *We claim no novelty for the irrationality of φ itself* (classical; already in Mathlib via $\sqrt{5}$): this is an independent Casoratian route and a demonstration witness that the pipeline closes an irrationality theorem, in the spirit of the by-hand faithfulness witnesses (Proposition 6), not a new number-theoretic result — see Remark 10.
- **VERIFIED** (numerically/symbolically; Section 8): the law at $k = 3$ symbolically and for $k = 2, \dots, 7$ by exact-rational checks, plus the named instances.
- **CONJECTURED / open**: a concrete simultaneous-Hermite–Padé PCF whose order-3 Casoratian drives the now-proven pipeline (Section 6) to the irrationality of a constant *not already known* to be irrational — an Apéry-style certificate ($\zeta(3)$ -type) rather than the golden-ratio demonstration. This is the genuinely open next entry: the determinantal machinery, the converse, and the certificate engine it consumes are now all in place; what remains is to exhibit a target recurrence whose shrinking remainder yields a *new* irrationality theorem.

We claim no novelty for Abel’s formula itself, which is classical ([1, Ch. 2],[2],[3]), nor for the irrationality of the golden ratio; the contributions are the uniform-in- k machine-checked proof, the formalized fundamental-system converse, the reusable machine-checked irrationality pipeline, and the opening of the order- ≥ 3 direction in this corpus.

10 Relation to prior work and novelty

Formal-methods landscape. The two standard formal-library treatments of linear recurrences both target the *constant-coefficient* solution theory, not the Casoratian evolution law. Mathlib’s `LinearRecurrence` [6] fixes an order and a tuple of *constant* coefficients `coeffs : Fin order  $\rightarrow \mathbb{R}$` , builds the solution submodule of $\mathbb{N} \rightarrow R$, the `LinearEquiv` sending a solution to its first `order` terms (so solutions are pinned by their initial data), and the characteristic polynomial; it defines no Casoratian and proves no Abel-type evolution law. Eberl’s *Linear Recurrences* entry in the Isabelle Archive of Formal Proofs [9] develops constant-coefficient recurrences via formal power series and polynomial factorisation into an executable closed-form solver, again without the Casoratian/Abel identity. Our object is complementary and, in one direction, strictly more general: the coefficients $c_j(n)$ are arbitrary functions of n (time-varying), the Casoratian is a genuine `Matrix` determinant, and the theorem is the determinantal evolution law uniform in the order k — exactly the non-vanishing / linear-independence certificate (Corollary 5) that Hermite–Padé and Apéry-type constructions consume ([4, 5]). A machine-checked Casoratian does already exist for the order-2 case in another system: the Coq/MathComp formal proof of the irrationality of $\zeta(3)$ [10] defines the 2×2 Wronskian `ba_casoratian` of the two Apéry sequences and derives its first-order (Abel) law, but only for that concrete instance — indeed its source remarks that the order-1 recurrence “should be obtained from something more general.” Having searched Mathlib, the Isabelle AFP, and the Coq/MathComp ecosystem by name (*Casoratian*, *Wronskian*) and by structure (as of June 2026), we find no *machine-checked* general-order, uniform-in- k Casoratian Abel formula with time-varying coefficients in any of these systems — precisely the “something more general” the Apéry formalization lacked. The *identity* itself is of course classical and not in question: the general-order law with time-varying coefficients is textbook ([1, Ch. 2],[2],[3]), for which we claim no novelty. Our claim is strictly one of *formalization coverage* — the machine-checked theorem, uniform in k and with time-varying $c_j(n)$, that the present note contributes and gate-checks. The irrationality pipeline of Section 6 is likewise complementary to [10]: where that development hard-wires a 2×2 Casoratian into a single bespoke $\zeta(3)$ certificate, ours is a *reusable* order- $(k+1)$ engine that consumes any integer fundamental set and a shrinking real remainder, here demonstrated on the classical golden ratio (for which we claim no number-theoretic novelty, Remark 10) and awaiting a target not already known irrational (Section 9).

Within this corpus, and a note on hosting. The deposited corpus had formalized the Casoratian only in its order-2 form — the $4/\pi$ PCF Casoratian closed form (Theorem 6.3 of the ladder paper [12]; machine-verified in [11]) and the general order-2 `casoratian_eqG` in `pcf-delta` [13] — so this is the program’s first order- ≥ 3 result and the backbone of a planned Hermite–Padé / Apéry line. We deposit it as a standalone catalogue entry rather than as a Mathlib pull request for two reasons: the result is more general than Mathlib’s constant-coefficient `LinearRecurrence` API and would need new time-varying-coefficient scaffolding to land there, and the corpus requires the core to be hosted and cone-gated inside `pcf-delta`’s `Check.lean`. A Mathlib contribution — a `Casoratian` determinant and its Abel evolution law, specialising to `LinearRecurrence` in the constant-coefficient case — is a natural next step and would convert “not otherwise available” into

an externally reviewed library lemma. For the record, the Lean and `sympy` sources label this identity “Abel–Jacobi–Liouville”; the standard discrete name, used here, is Abel’s formula (Elaydi [1]), the triple form naming the continuous-Wronskian analogue (Liouville / Jacobi). The note supplements the SIARC program [14].

11 Code and data availability

The Lean sources are `verify/GeneralCaso.lean` (namespace `PcfGeneralCaso`; the uniform-in- k law and the fundamental-system converse), `verify/HigherCaso.lean` (namespace `PcfHigherCaso`; the by-hand low-order witnesses), and `verify/HermitePade.lean` (namespace `PcfHermitePade`; the irrationality pipeline and the concrete order-3 certificates); they are built and axiom-cone-checked inside the `pcf-delta PcfContinuant` project [13] via `lake env lean Check.lean` (toolchain `leanprover/lean4:v4.30.0`, `Mathlib rev=v4.30.0`); the verbatim `#print axioms` output for the twenty-six declarations is mirrored in this repository as `verify/AXIOM_CONES.txt`, so the cone partition of Section 7 is reproducible here without rebuilding the host project. The independent symbolic gate is `harness_caso_k.py` (`sympy` [8]; run `python harness_caso_k.py`), which prints the $k = 3$ symbolic proof, the exact-rational $k = 2, \dots, 7$ verification, and the named instances. Public repository: <https://github.com/papanokechi/pcf-casoratian-catalogue>.

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